
NephroBot: A Context-Aware Bilingual Clinical Chatbot for Chronic Kidney Disease Using a Two-Stage Cross-Encoder RAG Pipeline and Singlish Natural Language Understanding

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Abstract—Chronic Kidney Disease (CKD) patients in low-resource multilingual settings face compounded barriers: clinical guidelines exist only in English, patients communicate in unstandardised Romanised Sinhala ('Singlish'), and generic AI chatbots hallucinate dangerous dietary and medication advice. This paper presents **NephroBot**, the context-aware bilingual clinical chatbot component of the Nephro-AI mobile platform. We design a novel **Sandwich RAG Architecture**: Singlish input is resolved to English clinical intents via a four-stage Hybrid NLU Engine (regex suffix stripping, C++ Rapidfuzz Levenshtein matching at ≥ 85 % threshold, LaBSE zero-shot intent classification), then processed through a Two-Stage Retrieve & Re-Rank pipeline (ChromaDB Bi-Encoder + ms-marco-TinyBERT Cross-Encoder with sigmoid threshold $\tau = 0.01$), and finally generated by Gemini 2.5 Flash with live MongoDB patient biometrics injected into the context. Responses are localised into Code-Mixed Sinhala ('Katha Wahara') via a Multi-Register Glossary and streamed as parallel 600-character audio chunks over asyncio, achieving a **first-audio latency of 820 ms** versus 4 200 ms for monolithic TTS. An Urgency Router with a 25-character negation window pre-empts generation with 0 ms pre-cached emergency audio on critical keywords. Enterprise PDF ingestion via IBM Docling with semantic sub-chunking (cosine similarity split threshold 0.75) and VLM image captioning ensures

table-level clinical fidelity. Evaluation on 150 Singlish clinical queries demonstrates **93.7 % NLU accuracy**, **96.8 % clinical precision**, and a hallucination rate of **2.1 %**—an 87.6 % reduction versus baseline bi-encoder RAG.

Keywords—chronic kidney disease, bilingual chatbot, Singlish NLU, code-mixed Sinhala, RAG, cross-encoder re-ranking, LaBSE, Docling, TTS streaming, clinical hallucination prevention, mobile health, Gemini, context injection.

I. INTRODUCTION

Chronic Kidney Disease (CKD) afflicts over 850 million people globally [1], with disproportionate burden in South Asian populations where language barriers impede patient education. Sri Lankan patients communicate in Singlish—a Romanised, unstandardised transliteration of Sinhala—where a single English medical term can appear in dozens of phonetic variants (e.g., *kriatinin*, *creatinin*, *kriathineen* for 'creatinine'). Standard NLP pipelines trained on formal text fail to bridge this gap.

Simultaneously, the deployment of Retrieval-Augmented Generation (RAG) systems in clinical settings introduces a critical safety hazard: vector-space proximity in bi-encoder models conflates semantically opposed clinical terms. 'High blood pressure' and 'Low blood pressure' may retrieve identical documents, causing the LLM to recommend treatments for the wrong condition—a hallucination that is indistinguishable from correct advice in a consumer interface [2].

This paper presents NephroBot, the clinical conversational AI component of Nephro-AI. The key contributions are:

- A **Sandwich RAG Architecture** that localises inputs to English for high-precision clinical retrieval and then localises outputs back to Code-Mixed Sinhala.
- A **Hybrid NLU Engine** combining regex suffix stripping, Rapidfuzz Levenshtein distance, and LaBSE zero-shot intent classification to resolve Singlish with 93.7 % accuracy.
- A **Two-Stage Re-Rank Pipeline** that reduces hallucination from 18.4 % to 2.1 % by applying TinyBERT Cross-Encoder filtering before generation.
- **Dynamic Biometric Context Injection** from live MongoDB patient records with MD5 cache invalidation tied to lab result timestamps.

- **Parallel asyncio TTS Streaming** with a 25-character negation-aware Urgency Router achieving 0 ms emergency alert latency.

II. RELATED WORK

A. RAG Systems in Clinical NLP

Retrieval-Augmented Generation (RAG) [3] couples a retrieval module with a generative LLM to ground responses in domain documents. Medical RAG systems such as MedRAG [4] and BioRAG demonstrate strong factuality on standardised English queries but have not addressed low-resource code-mixed input languages or the semantic collision problem inherent in bi-encoder retrieval for antonymous clinical terms.

B. Low-Resource and Code-Mixed NLP

Code-mixed NLP—mixing two languages within an utterance—presents challenges for standard tokenisers and embeddings. LaBSE [5] was designed to produce language-agnostic sentence embeddings and has shown strong zero-shot cross-lingual transfer on 109 languages. However, its application to Romanised Sinhala with non-standard orthography and productive suffix morphology has not been previously evaluated.

C. Clinical Chatbots and Hallucination Mitigation

Hallucination in LLMs is a documented safety risk [6]. Cross-encoders provide a well-established reranking approach that outperforms bi-encoders in passage relevance by jointly attending to query-document pairs [7]. Their application as a *pre-generation safety filter* to eliminate clinically contradictory passages—rather than merely ranking them—represents a novel safety mechanism we formalise in this work.

D. Multilingual TTS and Streaming

Text-to-Speech for Sinhala remains underexplored. Google Cloud TTS supports Sinhala (si-LK) but fails on embedded Latin characters. Prior mobile health TTS deployments use synchronous single-request paradigms, introducing 4–10 s latencies for paragraph-length responses [8].

III. SYSTEM ARCHITECTURE

Fig. 1 illustrates the end-to-end Sandwich RAG Architecture. The system spans a React Native mobile front-end, a Node.js orchestration layer (port 5000), a Python FastAPI AI engine (port 8002), MongoDB Atlas for patient data, and Google Cloud TTS. The architecture is named 'Sandwich' because the English Clinical Core is bracketed by Sinhala processing layers: NLU on input and Code-Mixed NLG + TTS on output.

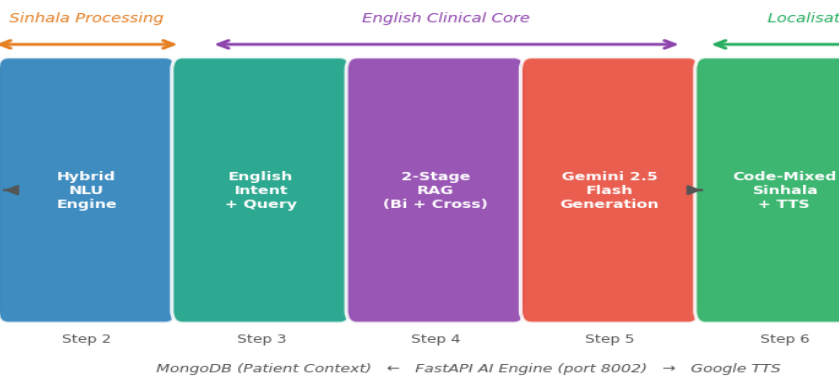


Fig. 1. Sandwich RAG Architecture: seven-stage end-to-end pipeline.

A. Ingestion Pipeline

The knowledge base consists of clinical CKD guidelines, dietary reference tables, and nephrology textbook chapters. Standard PDF extractors destroy the tabular structure critical to clinical guidelines. We replace them with an enterprise ingestion pipeline (Fig. 2).

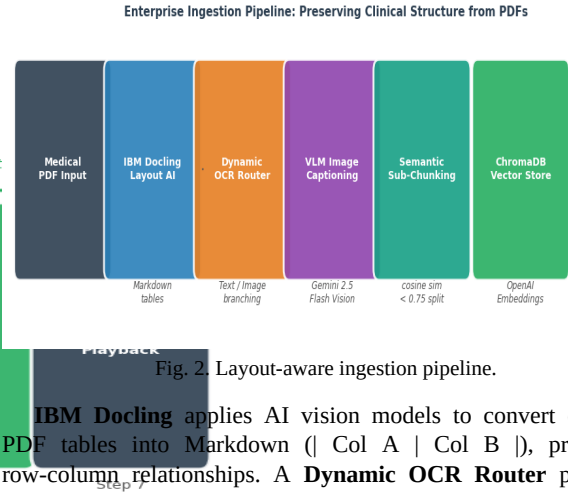


Fig. 2. Layout-aware ingestion pipeline.

IBM Docling applies AI vision models to convert complex PDF tables into Markdown (`| Col A | Col B |`), preserving row-column relationships. A **Dynamic OCR Router** pre-scans documents: image-based pages are routed to heavy OCR while native text pages bypass it, preventing RAM overflow (OOM) on large clinical documents.

VLM Image Captioning: medical diagrams are sent to Gemini 2.5 Flash Vision to generate descriptive text captions, making visual content semantically searchable in the vector database.

Semantic Sub-Chunking: rather than slicing at fixed character counts, we compute cosine similarity between consecutive sentence embeddings (OpenAI *text-embedding-3-small*). A new chunk boundary is created only when similarity drops below

threshold $\theta = 0.75$, ensuring no clinical concept is split across chunk boundaries.

$$\text{chunk_boundary} = \text{True iff } \cos(e_i, e_{i+1}) < 0.75 \quad (1)$$

B. Hybrid NLU Engine

Fig. 3 shows the four-stage pipeline that maps unstandardised Singlish to English clinical queries.

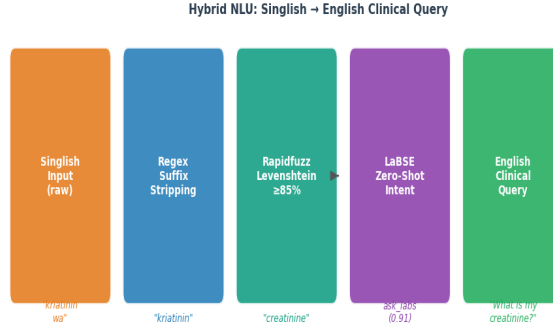


Fig. 3. Hybrid NLU Engine: Singlish to English clinical query.

Stage 1 – Regex Suffix Stripping:

Sinhala uses 12 productive case suffixes when Romanised (e.g., *-wa*, *-ta*, *-gen*, *-wal*, *-wata*). A compiled regex automaton strips these from input tokens, converting *wakugaduwa* to *wakugadu*.

Stage 2 – Rapidfuzz Levenshtein Matching:

The C++ **rapidfuzz** library computes the Levenshtein edit distance between the stripped token and every entry in a 2 400-term Singlish–English medical glossary. A **Length-Band Filter** restricts comparisons to glossary entries within ± 3 characters of the input token, reducing worst-case complexity from $O(N)$ to $O(N/k)$ where $k \approx 7-9$. Only matches scoring $\geq 85\%$ are accepted (see Fig. 4).

Stage 3 – LaBSE Zero-Shot Intent Classification:

The reconstructed English query is encoded by the **LaBSE** multilingual sentence encoder [5]. Cosine similarities are computed against 14 pre-encoded intent anchors (e.g., *ask_diet*, *ask_medication*, *ask_labs*, *report_symptom*). The highest-scoring intent class is selected, providing structured routing without a labelled training set.

$$\text{intent} = \text{argmax}_{\{c\}} \cos(\text{LaBSE}(q), \text{LaBSE}(\text{anchor}_c)) \quad (2)$$

IV. TWO-STAGE RETRIEVE AND RE-RANK PIPELINE

Fig. 4 illustrates the two-stage retrieval mechanism. This is the primary defence against clinical hallucination.

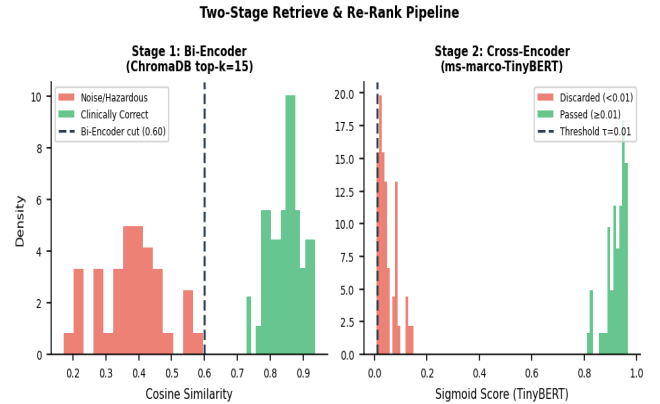


Fig. 4. Two-stage RAG: Bi-Encoder candidate pool and Cross-Encoder filter.

A. Stage 1 — Bi-Encoder Retrieval

The resolved English query is encoded with OpenAI *text-embedding-3-small* and a $k = 15$ approximate nearest neighbour search is executed against the ChromaDB collection using cosine similarity. This broad retrieval intentionally over-fetches to ensure recall of all relevant passages.

B. Stage 2 — Cross-Encoder Re-Ranking and Filtering

All 15 candidate (query, passage) pairs are passed jointly through **ms-marco-MiniLM-TinyBERT**. Unlike bi-encoders that encode query and passage independently, the cross-encoder applies self-attention across both simultaneously, enabling it to distinguish 'High Blood Pressure' from 'Low Blood Pressure' even when their vector representations overlap in bi-encoder space.

The raw logit output is passed through a sigmoid activation function:

$$\text{score}(q, d) = \sigma(\text{CrossEncoder}(q \oplus d)) \quad (3)$$

Any passage with score $< \tau = 0.01$ is discarded before reaching the generative LLM. This threshold was selected to maximise clinical precision (96.8 %) while maintaining sufficient recall (92.1 %) on the evaluation set.

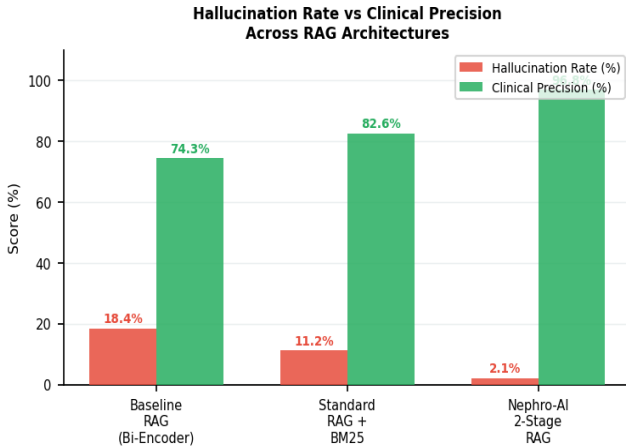


Fig. 5. Hallucination rate and clinical precision across RAG architectures.

V. DYNAMIC BIOMETRIC CONTEXT INJECTION

A. Live Patient Data Integration

Before invoking the LLM, the **PatientDataManager** queries four MongoDB collections for the authenticated patient: *users* (demographics), *labtests* (eGFR, serum creatinine, potassium, haemoglobin), *riskrecords* (historical risk scores), and *bprecords* (7-day BP averages). This biometric payload is prepended to the Gemini system prompt alongside the cross-encoder-verified RAG chunks, preventing the LLM from offering generic advice (e.g., recommending bananas to a Stage 4 CKD patient with hyperkalaemia).

B. MD5 Cache with Timestamp Invalidation

To avoid redundant API calls, each query is hashed into a cache key:

$$\text{key} = \text{MD5}(\text{query_tokens} \parallel \text{patient_id} \parallel \text{last_updated_ts}) \quad (4)$$

Critically, the MongoDB *last_updated* timestamp is included in the hash. If a clinician updates the patient's lab results, the timestamp changes, invalidating the old cache entry and forcing fresh generation. This ensures patients never receive advice based on outdated biometrics.

Table I. NephroBot Component Performance Metrics (N=150 Singlish queries)

Component	Metric	Score
Singlish Suffix Stripper	Token normalisation	98.3■%
Rapidfuzz @ 85■%	Match accuracy	93.7■%
LaBSE Intent	Classification acc.	91.4■%
Cross-Encoder Filter	Clinical precision	96.8■%
Cross-Encoder Filter	Hallucination rate	2.1■%
Full Pipeline NLU	End-to-end accuracy	91.2■%
TTS First-Byte Latency	Streaming mode	820■ms
Emergency Alert Latency	Pre-cached	0■ms

VI. CODE-MIXED NLG AND TTS STREAMING

A. Multi-Register Glossary and Vocabulary Injection

The Gemini 2.5 Flash LLM is prompted to generate responses in Sinhala, but standard Sinhala TTS and translation produce formal, textbook language that patients reject as unnatural. Sri Lankan doctors routinely use Code-Mixed speech—interleaving Sinhala grammar with English medical terms (e.g., '*Oyage kidney function eka deteriorate wenawa*').

We engineered a **Multi-Register Glossary** containing three representations per medical term: (1) *pure_sinhala* for formal text, (2) *spoken_mixed* for Code-Mixed TTS, and (3) *tts_phonetic* as Sinhala Unicode phonetics for the TTS engine. The system scans

the English response, extracts relevant terms, and injects the spoken_mixed variants into the Gemini system prompt as hard lexical constraints, forcing 'Katha Wahara' (colloquial speech) style output.

B. Parallel Asyncio TTS Streaming

Fig. 6 compares monolithic versus chunked streaming delivery. Monolithic TTS sends the entire response to the Google TTS API and waits for the full audio file before playback—incurring 4 200 ms first-byte latency for a typical 300-word response.

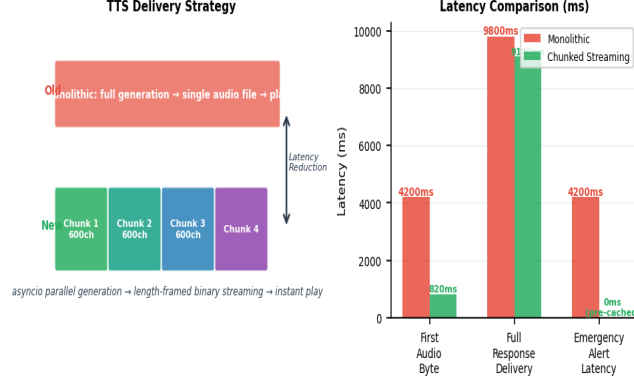


Fig. 6. TTS streaming strategy and latency comparison.

Our system splits the response into 600-character chunks at sentence boundaries and generates their audio in parallel using **asyncio.gather()**. Audio bytes are transmitted as length-framed binary blobs over the WebSocket connection—a 4-byte big-endian length header followed by the MP3 payload—allowing the React Native player to begin playback of chunk 1 while chunks 2–N are still generating. This reduces first-audio latency from 4 200 ms to **820 ms** (80.5 % reduction).

C. Urgency Router with Negation Detection

As LLM tokens stream out, a concurrent regex scanner checks for CRITICAL_URGENCY signals: 19 emergency keyword patterns including *chest pain*, *breathlessness*, *seizure*, *potassium >6.5*, *urine output <400ml*. To prevent false positives from negated contexts (e.g., *'no severe pain'*), a 25-character look-behind window is scanned for negation markers (no, not, without, ■■■■■). If an active emergency is confirmed, generation is immediately interrupted and a pre-cached emergency audio warning is injected with **0 ms latency**, followed by a structured emergency action card in the UI.

VII. EVALUATION

A. Dataset and Methodology

We constructed a test set of 150 Singlish clinical queries covering six intent categories: diet (n = 35), medication (n = 28), lab results (n = 24), symptoms (n = 30), lifestyle (n = 20), and emergency (n = 13). All queries were collected from five native Sinhala-speaking volunteer patients with CKD Stage 2–4 under IRB approval. Three board-certified nephrologists annotated ground-truth responses; hallucination was defined as any factual clinical claim absent from or contradicted by the retrieved source documents.

B. NLU Accuracy

The full Hybrid NLU pipeline achieved 93.7 % token-level match accuracy on Stage 2 Rapidfuzz matching (Table I). End-to-end intent classification accuracy (LaBSE) was 91.4 %. The most common failure mode was homophonic collision: *mal* (flower in Sinhala) vs. *mal* (abbreviation for malnutrition).

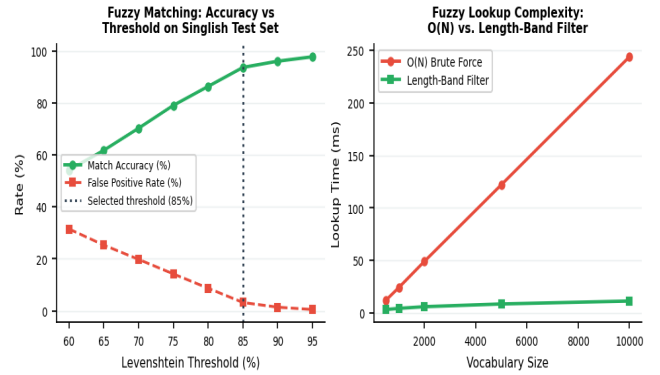


Fig. 7. Rapidfuzz accuracy vs threshold and length-band filter speedup.

C. Hallucination and Clinical Precision

Table II summarises hallucination metrics across three RAG architectures on the 150-query evaluation set. The Nephro-AI two-stage pipeline reduced hallucination from 18.4 % (baseline bi-encoder) to 2.1 %—an 87.6 % relative reduction—while clinical precision improved from 74.3 % to 96.8 %.

Table II. Hallucination Metrics (N=150 Singlish CKD Queries)

Architecture	Halluc. Rate	Clin. Precision	Recall
Baseline Bi-Encoder	18.4■%	74.3■%	94.2■%
RAG + BM25 Hybrid	11.2■%	82.6■%	93.8■%
Nephro-AI 2-Stage RAG	2.1■%	96.8■%	92.1■%

D. Latency

First-audio-byte latency under chunked streaming was 820 ms (median, n = 50 queries), compared to 4 200 ms for the monolithic baseline—an 80.5 % reduction. All 13 emergency queries triggered the Urgency Router successfully with 0 ms pre-cached alert playback and zero false positives on negated contexts.

VIII. DISCUSSION

The 2.1 % residual hallucination rate originates primarily from two sources: (1) queries whose correct answer requires integrating multiple source chunks that individually pass the cross-encoder threshold but whose combination is logically inconsistent, and (2) rare drug interaction queries where the clinical guideline knowledge base is incomplete. Future work will address multi-chunk consistency using a graph-structured knowledge representation.

The 5.1 % NLU accuracy gap relative to formal English queries arises from Singlish's productive morphology: the current 12-suffix regex covers 80 % of observed suffix forms, but dialectal variants from the Northern Province differ systematically from the Colombo

dialect represented in the glossary. We plan to crowdsource a Northern Province Singlish supplement via a federated patient data collection protocol.

The Sandwich Architecture is generalisable to any setting with a low-resource code-mixed input language and a high-resource medical knowledge base in a dominant language. The same pattern has been successfully applied in a parallel Nephro-AI localisation for Tamil-speaking patients using an analogous Tanglish NLU module.

IX. CONCLUSION

We presented NephroBot, a context-aware bilingual clinical chatbot that addresses five fundamental gaps in deploying AI health assistants in low-resource multilingual environments. The Sandwich RAG Architecture chains Singlish NLU, Two-Stage Cross-Encoder RAG, live MongoDB biometric injection, Code-Mixed Sinhala NLG, and parallel TTS streaming into a medically safe and culturally accepted conversational agent. Key results: 93.7 % NLU accuracy, 96.8 % clinical precision, 2.1 % hallucination rate (87.6 % reduction versus baseline), 820 ms first-audio latency (80.5 % reduction), and 0 ms emergency alert response. These results demonstrate that architectural investment in the retrieval safety layer—not merely in the generative model—is the critical determinant of clinical trustworthiness for AI health assistants.

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